MILITARY SPECIFICATION SHEET

VOLTMETER, DC, AIRCRAFT, DIGITAL DISPLAY INTEGRAL LIGHTED, 25 4 MILLIMETERS

This specification sheet is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the voltmeter described herein shall consist of this specification and the latest issue of MIL-V-85603

Requirements

Military part number Military part numbers of the voltmeters shall be as follows

Military Part Number	Type of Voltmeter
D85603/3-1	White integrally lighted (5 0VAC 400Hz)
D85603/3-2	"NVIS Green A" integrally lighted (5.0VAC 400Hz)

Each voltmeter delivered shall be supplied with four appropriate screws and positive locking devices (washers) and each voltmeter shall be equipped with four press seal of equivalent electrical feed through connections in accordance with Figure 1 herein.

Caution advisory The caution advisory switch is not required for this application

Low voltage warning The low voltage warning threshold voltage is 21 0 volts

Integral lighting The voltmeter (D85603/3-1) shall be integrally lighted Instrument and Panel Lighting (IPL) white, and the color shall be in accordance with MIL-P-7788 The lighting circuitry and brightness shall be in accordance with the requirements of MIL-L-25467 Voltmeter (D85603/3-2) shall be Night Vision Imaging System (NVIS) compatible in accordance with the requirements of MIL-L-85762 The voltmeters shall exhibit a contrast ratio **(C_L) of not less than 5 when tested over the range of 1.0 foot-candle to 10,000 foot-candles.**

AMSC N/A **FSC 6625**

Distribution Statement A **Approved for public release; distribution is unlimited.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

Dimensions. The voltmeter shall conform to the dimensional and display requirements of Figure 1 herein.

Weight. The weight of the voltmeter when fully assembled and configured for in-service use shall be not greater than 0 23 kilogram.

Methods of inspection

Electrical performance. The voltmeter shall be inspected to determine compliance with the electrical performance requirements of 3 5 1 as follows Each voltmeter shall be connected to a source of power and the input voltage varied over the range of 15 volts to 35 volts The voltmeter shall display the input voltage to within ± 0.10 volt over the temperature range of +71°C to -32°C, and display the input voltage to within ± 0 20 volt at temperatures between -37°C to -54°C The voltmeter readings shall be recorded in 1-volt intervals between 15 and 35 volts With the input voltage to the voltmeter adjusted above 21 volts but not exceeding 60 volts, the low voltage warning display shall not be visible. The input voltage to the voltmeter shall be gradually reduced to 21 volts and at this point, the low voltage warning display shall be visible and exhibit the following duty cycle. blink at a rate of 4 ± 1 time(s) per second for 1 ± 0 10 minute, and remain activated in a steady state condition for an additional 10 ± 1 minute(s) The digital low voltage warning display shall continue to repeat the duty cycle until the low voltage condition no longer exists or the voltmeter input voltage degrades to the point where the voltmeter becomes inoperable

Vibration The voltmeter shall operate as prescribed when required during vibration testing and there shall be no evidence of mechanical or electrical degradation when the voltmeter is tested after the completion of the vibration tests The voltmeter shall be subjected to the vibration tests of MIL-STD-810, Method 514.3, Procedure I, Test Condition I-3.2 12; MIL-STD-810, Method 514 3, Procedure I, Test Condition I-3 2 5, MIL-STD-810, Method 519 3, Section II, Procedure II-3 and Figures 3, 4, and 5 of this specification, respectively

Sinusoidal (monitored) MIL-STD-810, Method 514 3, Procedure I, Test Condition I-3 2 12 and Figure 3 The test time for each of the three mutually perpendicular axis shall be 3 hours for a total of 9 hours Degradation of the voltmeter's electrical or mechanical performance shall be cause for rejection

Random non-gunfire (unmonitored). MIL-STD-810, Method 514 3, Procedure I, Test Condition I-3 2 5 and Figure 4 The test time for each of the three mutually perpendicular axis shall be 2 hours and 30 minutes Degradation of the voltmeter electrical performance shall be permitted, but degradation of the voltmeter's mechanical performance requirements shall be cause for rejection

Random non-gunfire (monitored). MIL-STD-810, Method 514.3, Procedure I, Test Condition I-3.2.5 and Figure 4 The test time for each of the three mutually perpendicular axis shall be 30 minutes After 2 hours and 30 minutes have elapsed as specified in (unmonitored), the vibration level of 7.4 grms shall be reduced to 3.4 grms and the voltmeter shall be monitored for an additional 30 minutes. Degradation of the voltmeters electrical or mechanical performance shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

Random gunfire (unmonitored) MIL-STD-810, Method 519 3, Section II, Procedure II-3 and Figure 5 The test time for each of the three mutually perpendicular axis shall be 30 minutes Degradation of the voltmeter's electrical performance shall be permitted but any degradation of the voltmeter's mechanical performance shall be cause for rejection At the conclusion of the above tests, the voltmeter shall meet the requirements herein and of DOD-V-85603

Electromagnetic interference The voltmeter shall be tested in accordance with MIL-STD-462 for those tests required by MIL-STD-461 for Class A1, Category A1b equipment The failure of the voltmeter to meet any of the requirements of the following tests shall be cause for rejection CE01, CE03, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03 except that the test levels for RS03 shall be as follows

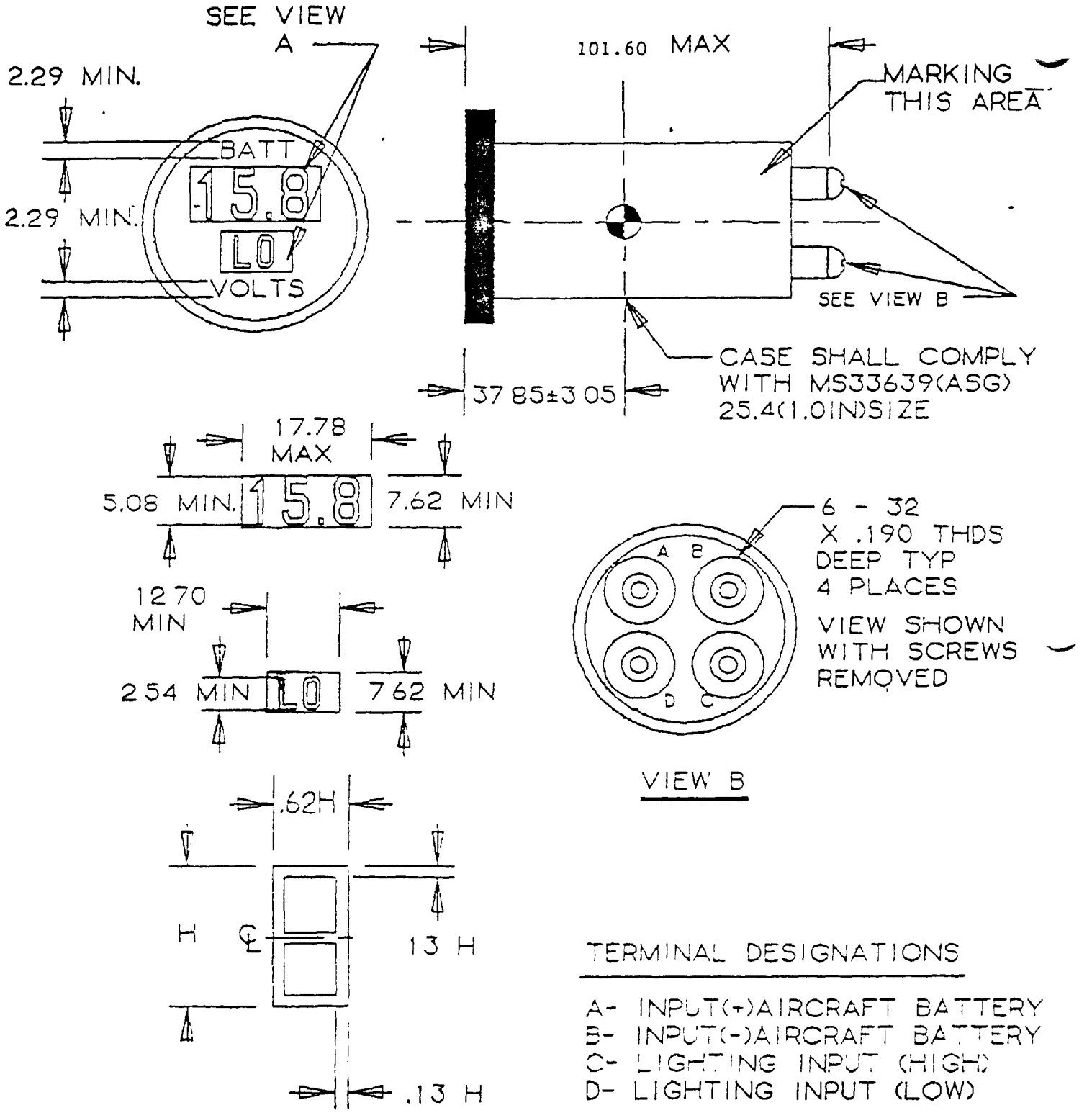
E-Field	Frequency Range
20V/meter	14 KHz to 200 MHz
65V/meter	200 MHz to 450 MHz
20V/meter	450 MHz to 1 0 GHz
100V/meter	1 0 GHz to 12 0 GHz
20V/meter	12 0 GHz to 40 0 GHz except 16 5 GHz and 33 0 GHz
100V/meter	16 5 GHz and 33 0 GHz

Integral lighting The voltmeter (D85603/3-1) shall be integrally lighted Instrument and Panel Lighting (IPL) white, and the color shall be in accordance with MIL-P-7788 The lighting circuitry, brightness, and stray light shall be in accordance with the requirements of MIL-L-25467 Voltmeter (D85603/3-2) shall be tested in accordance with MIL-L-85762 to insure compatibility with Night Vision Imaging System (NVIS) The voltmeters shall not exhibit a contrast ratio (C_R) of less than 5 over the range of 1.0 foot-candle to 10,000 foot-candles During these tests, the voltmeter shall be energized and the input voltage adjusted to 15 8 VDC When required, the lighting circuit shall be energized with the appropriate voltage

Dial Substitute BATT VOLTS for DC VOLTS

Preparing Activity
Navy - AS
(Project No 6625-N973-03)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.



TERMINAL DESIGNATIONS

A- INPUT(+)AIRCRAFT BATTERY
B- INPUT(-)AIRCRAFT BATTERY
C- LIGHTING INPUT (HIGH)
D- LIGHTING INPUT (LOW)

DIMENSIONS IN MILLIMETERS. TOLERANCE UNLESS OTHERWISE SPECIFIED: XX± .76

FIGURE 1- Aircraft digital voltmeter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

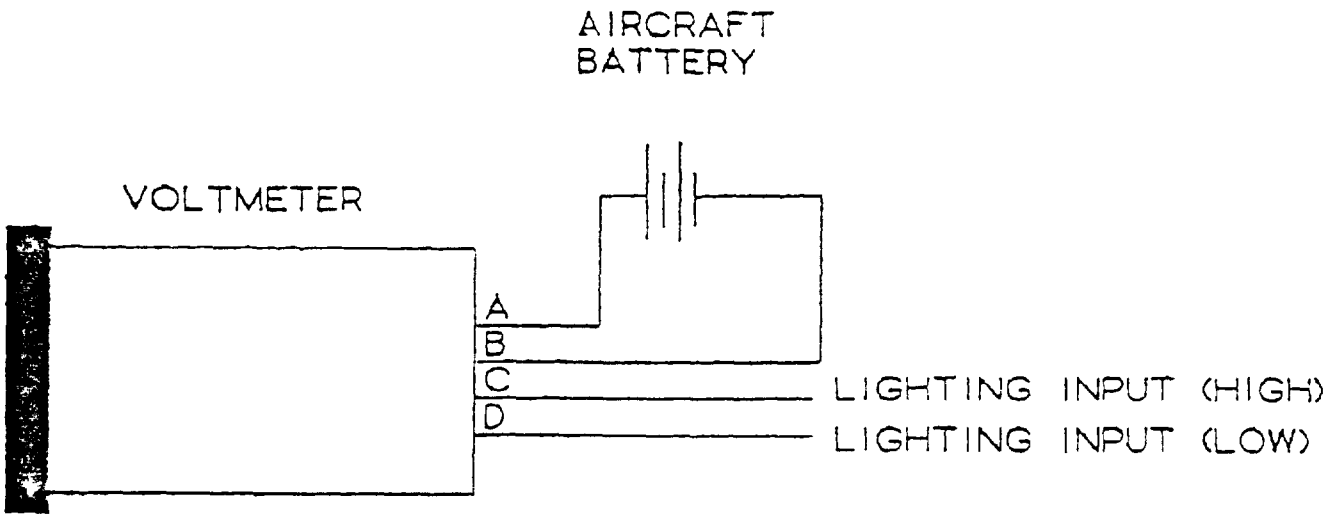


FIGURE 2. External circuitry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

SINUSOIDAL TEST SCHEDULE

(Test times in minutes per axis)

ORDER	# OF EQUIPMENT RESONANCES	0	1	2	3	4
1	Total Dwell Time at Equipment Resonances	0	30	60	90	120
2	Total Cycling Time (5-2000-5 Hz 20 min. log sweep)	180	150	120	90	60
Total Time Each Axis		180	180	180	180	180

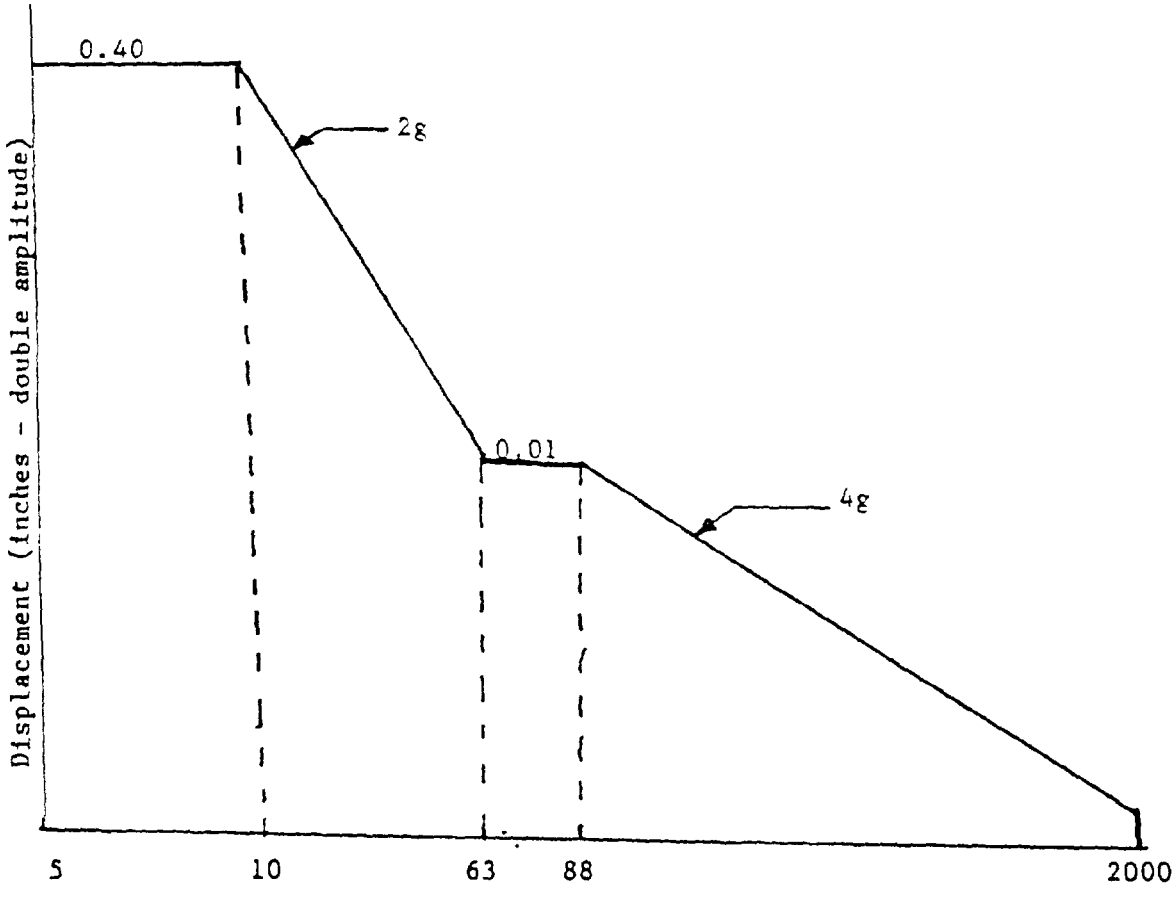
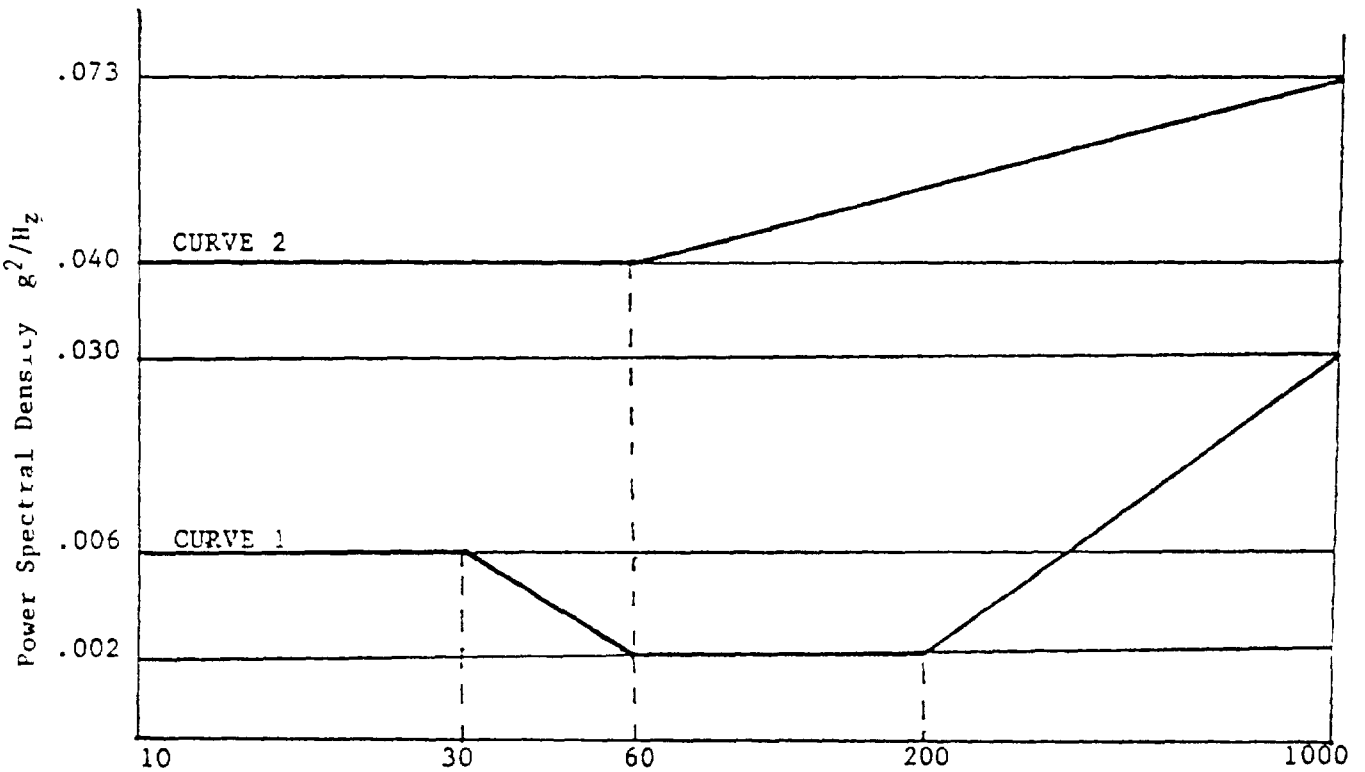


FIGURE 3 - Vibration sinusoidal test requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

NON-GUNFIRE

(Test time - 3 hours per axis)



Frequency - (Hz)

CURVE 1 monitored composite - 3.4 grms

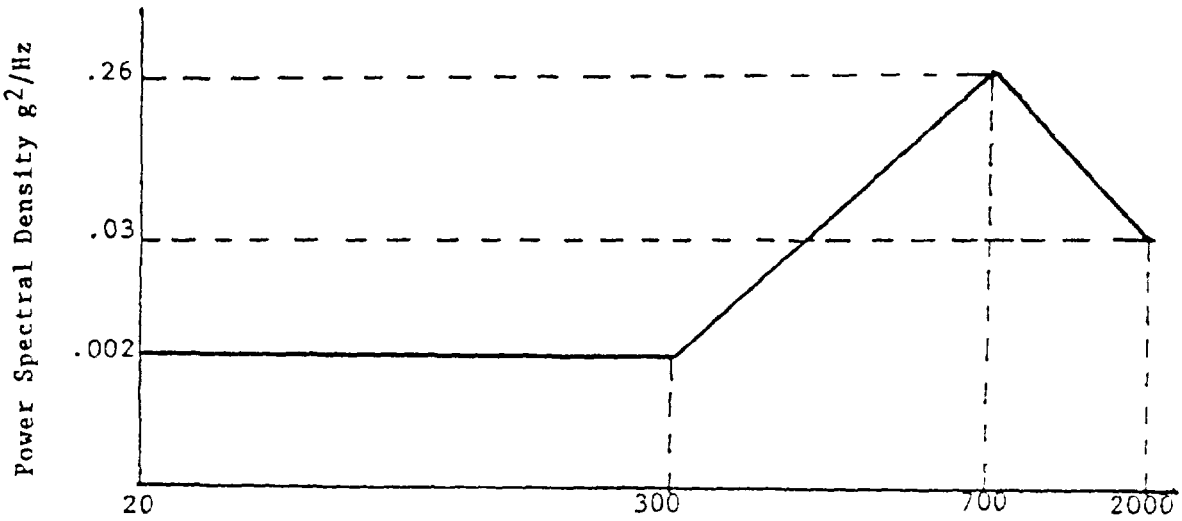
CURVE 2 unmonitored composite - 7.4 grms

FIGURE 4 - Vibration random test requirements, (non-gunfire).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

GUNFIRE

(Test time - 30 minutes per axis)



GUNFIRE composite - 12.0 grms

FIGURE 5 - Vibration random test requirements, (gunfire).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.